

Frozen Theorem Statements: Simple Critical-Line Zeros

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15 September 2026
Corrected public theorem sheet, v0.1.1

Status

This note extracts theorem-level statements from the frozen post-Gebendorfer fusion proof. The frozen exact author verifier returns `PASS_EXACT`; the hostile audit passed K1–K15; the final publication referee returned `PUBLICATION_PASS_WITH_EDITORIAL_DEFECTS_ONLY`. The Gebendorfer finite layer was independently cross-host replayed for 58,577,037 nodes. Lean formalization is partial. This v0.1.1 theorem sheet corrects public transcription/self-containment omissions identified by a post-release regression audit; the theorem endpoint and frozen proof are unchanged.

Notation

Let $N(T)$ count all nontrivial zeta zeros ρ with $0 < \text{Im } \rho \leq T$, with analytic multiplicity. Let $N_0^s(T)$ count distinct zeros $\rho = \frac{1}{2} + i\gamma$ with $0 < \gamma \leq T$ and multiplicity one. Put

$$\kappa_0^s = \liminf_{T \rightarrow \infty} \frac{N_0^s(T)}{N(T)}.$$

The frozen certified lower floor for the window constant $2 - C(f_0)$ used below is

$$h := \frac{210042647916503}{312500000000000}.$$

Theorem S1: simple critical-line zero proportion

$$\kappa_0^s \geq \frac{6734775921119}{10^{13}} = 0.6734775921119.$$

For the supplied certified Gebendorfer comparator

$$r_G = \frac{8269551442741204889710953}{12278882618209750000000000},$$

the exact gain is

$$\frac{6734775921119}{10^{13}} - r_G = \frac{34271814394247548721}{491155304728390000000000000} > 0.$$

Theorem S2: one common residual state

On the frozen normalized state,

$$x = s/N, \quad u = k/N, \quad v = \ell/N, \quad w = \beta/N, \quad c = q_0/N = 1 - x - 2w,$$

with $d = D/N$ and $y = Y/N$, one has

$$S_2 - 2N + s = D + Y, \quad hN + D + Y \leq s + o(N),$$

with

$$Y = 2q_0 + W + J + Z \geq 0.$$

The feasible region contains

$$h \leq x \leq 1, \quad 2u + v \leq x, \quad d + y \leq x - h,$$

$$d \geq u + 2v + \frac{(u + v)^2}{x - u}.$$

All later gains are restrictions or decompositions of this same residual budget.

Theorem S3: global residual payment

For the same normalized state and every $\rho \geq 0$ in the frozen route,

$$y - \rho v \geq -p_h(x; \rho),$$

where

$$p_h(x; \rho) = \frac{\sqrt{A_x^2 + \rho^2 N_x} - A_x}{2}, \quad A_x = (2x - h) + \rho x, \quad N_x = \frac{(2 - h)x - h}{2}.$$

Equivalently,

$$p_h^2 + A_x p_h = \frac{\rho^2 N_x}{4}.$$

Theorem S4: universal local supporting plane

For

$$m = 492, \quad \rho = \frac{19569717577}{10^{14}}, \quad R = \frac{38219880252651}{10^{13}},$$

and the six fixed nonnegative position prices

$$(t_0, \dots, t_5) = \left(\frac{2444609786511}{2500000000000}, \frac{28224041240321}{10^{14}}, \frac{24984358312191}{5000000000000}, \right. \\ \left. \frac{72302683681231}{10^{14}}, \frac{90853194486369}{10^{14}}, \frac{1249217915609}{1250000000000} \right),$$

put $n = m - 6 = 486$ and

$$B := \frac{1970462189}{500000000000}, \quad \tau := \frac{7}{6}.$$

For the m -point Gram block define

$$E := 2 \sum_{i < j, j-i \leq 6} |G_{ij}|^2, \quad e := \frac{E}{n}, \quad u_k^{\text{loc}} := \frac{BS_k}{n} \quad (0 \leq k \leq 5),$$

where S_k are the frozen gap-position sums used by the reflected-trim constraints. Define

$$G_{m,\rho}(E) = \begin{cases} E, & E \leq \tau(1 + \rho/2)^2, \\ \frac{E + m \left(2(1 + \rho/2)\sqrt{E/\tau} - (1 + \rho/2)^2 \right)}{1 + m/\tau}, & E > \tau(1 + \rho/2)^2. \end{cases}$$

The frozen tail-aware band profile then satisfies

$$G_{m,\rho}(ne) + n \sum_{k=0}^5 t_k u_k^{\text{loc}} \geq R$$

throughout the complete 69-row reflected-trim polyhedron. The certificate uses 464 exact rational chord/ray pieces with nonnegative Farkas weights; floating vertex completeness is not a premise.

Theorem S5: all-offset transport

$$\boxed{d + \rho v \geq \frac{Rx - P_{\text{price}}}{m}}, \quad P_{\text{price}} = \frac{47854801908769656526884823}{250000000000000000000000}.$$

Theorem S6: exact fusion separator

Let

$$s_0 = \frac{6734775921119}{10^{13}}.$$

For $h \leq x \leq s_0$, the frozen function $p_h(x; \rho)$ is increasing in x . The exact replay supplies a rational $\bar{p} \geq 0$ satisfying

$$\bar{p}^2 + A_{s_0} \bar{p} \geq \frac{\rho^2 N_{s_0}}{4}.$$

Since $p \mapsto p^2 + A_{s_0}p$ is increasing on $p \geq 0$, this gives $p_h(s_0; \rho) \leq \bar{p}$, and hence $p_h(x; \rho) \leq \bar{p}$ throughout $h \leq x \leq s_0$. It then verifies the strictly positive rational margin

$$M_{\text{end}} := mh - P_{\text{price}} - m\bar{p} - (m - R)s_0 > 0.$$

Its exact numerator and denominator are recorded in the frozen verifier and in the accompanying Markdown theorem sheet. Together with

$$d + y = (d + \rho v) + (y - \rho v),$$

this excludes $h \leq x \leq s_0$; the imported basic inequality excludes $x < h$.

Dependency map

certified local base $\longrightarrow$ common residual state $\longrightarrow$ 69-row plane,

69-row plane $\rightarrow$ transport + one global payment $\rightarrow$ exact separator.

Scope

This is not RH, not full Lean formalization, not completed human peer review, and not a literature-wide priority claim.